GLA University, Mathura

II-Mid Term Examination, 2011-12

Course: - B.Tech. II-Year, III-Sem.

Subject: - Mathematics III **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:- 40**

Note:-

1) Attempt **ALL** questions of Group A, **ANY TWO** from Group B and **ANY TWO** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

GROUP A (2 x 8 = 16)

Q1. Z-transform of U(k) is _____ where U (k) is unit step function.

Q2. Inverse Z-transform of $\frac{z}{z-a}$ is.....

Q3.The Fourier sine integral of the function F(x) is given by_____.

Q4. If f(p) is complex Fourier transform of the function F(x) then

$F\{F(x-a)\} = e^{iap} f(p)$ (True/False)

Q5. If f(k) = {1,2,3,4,5}.Then Z {f(k)} is_____.

Q6. If Z (y_k) = Y then Z{y_{k+4} } =.....

Q7. If f(p) is complex Fourier transform of the function F(x) then F{F(ax)}= _____

Q8. If f(k)*g(k) denotes the convolution of f(k) and g(k) and If Z{f(k)}= F(z) and Z{g(k)}= G(z) then Z{f(k)*g(k)}is equal to .....

GROUP B (Attempt **ANY TWO**) (5 x 2= 10)

Q1. Find inverse Z-transform of $F(z) = \frac{z}{z^2+7z+10}$

Q2. Using Fourier integral representation show that $\int_0^\infty \frac{w \sin wx}{1+w^2} dw = \frac{\pi}{2} e^{-x}$, $x > 0$

Q3. Find Z-transform of f(k) = sin αk , $k \geq 0$ and α is a constant.

GROUP C (Attempt **ANY TWO**) (7 x 2= 14)

Q1.Solve the difference equation : $y_{k+2} - 3\,y_{k+1} + 2\,y_k = 0$, $y_0 = 0$, $y_1 = 1$

Q2. Express the function $F(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$ as a Fourier integral. Hence evaluate

$$\int_0^\infty \frac{\sin\lambda \cos\lambda x}{\lambda} d\lambda$$

Q3. Find Fourier cosine transform of $F(x) = \begin{cases} x, & 0 < x < 1/2 \\ 1-x, & 1/2 < x < 1 \\ 0, & x > 1 \end{cases}$

GLA University, Mathura

II-Mid Term Examination, 2011-12

Course: - B.Tech. II-Year III-Sem. Date 14.11.11

Subject: -Environmental Studies **Univ. Roll No:-**

Time:-90 Minutes **Total Marks:-40**

Notes:-

1) Answer all questions from Group A (which is) **compulsory**, **Any Two** from Group B and **Any Two** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Group - A

Q. 1. Attempt all questions **(1 x 16)**

(i) The cycle in which main reservoir is the soil is called

(a) Biogeochemical cycle (b) Sedimentary cycle
(c) Sedimentary cycle (d) None

(ii) Extensive planting of tree to increase cover is called

(a) Afforestation (b) Agroforestation
(c) Defoerstation (d) Social forestry

(iii) Deforestation generally decreases

(a) Rainfall (b) Soil erosion
(c) Drought (d) Global warming

(iv) Which of the following is non-renewable resource?

(a) Coal (b) Forest
(c) Water (d) Wild life

(v) Fossil fuel and metallic minerals are

(a) Renewable resources
(b) Non-renewable resources
(c) Inexhaustible
(d) None of these

(vi) Maneb contains----------

(a) Mg (b) Mn (c) Hg (d) Pb

(vii) Chipko movement was initiated by

(a) Odom (b) Tanseley
(c) Whitteker (d) Sunder Lal Bahuguna

(viii) Which of the following energy is ecofriendly

(a) Hydropower (b) Fossil fuel
(c) Nuclear Power (d) Wood

(ix) Maximum tolerance limit of fluoride in drinking water is

(a) 0.5 ppm (b) 1.0 ppm (c) 1.5 ppm (d) 1.8 ppm

(x) Wind energy can not be used in

(a) Mountain areas (b) Dessert areas
(c) Forest areas (d) Coastal areas

(xi) Nuclear reaction taking place in hydrogen bomb

(a) Chemical reaction (b) Nuclear fusion
(c) Nuclear fission (d) Hydrogen reaction

(xii) Biomass energy can be obtained

(a) By energy plantation
(b) From petro crops
(c) From agricultural and urban wastes
(d) All of the above

(xiii) Sunlight may be converted into electricity through

(a) Galvanic cell (b) Photovoltaic cell
(c) Carbon electrode (d) Platinum electrode

(xiv) Fluorosis is caused by

(a) Deficiency of Fluoride
(b) Deficiency of fluorine
(c) Excess of Fluoride in potable water

(d) None

(xv) Renewable resources of energy include

(a) Solar energy (b) Natural gas

(c) Coal gas (d) Oil gas

(xvi) Which one is not a perpetual type of resource

(a) Wind energy (b) Rainfall

(c) Geothermal energy (d) Tides

Group -B

Q 2. Attempt any TWO questions (2 x 6)

(i) What is deforestation? Discuss various cause of deforestation.

(ii) What are natural resources? How will you classify the natural resources?

(iii) Write note on biogas as source of energy.

Group -C

Q 3. Attempt any TWO questions (2 x 6)

(i) What is material cycle? Explain it with the help of Nitrogen cycle.

(ii) What is solar energy? Discuss its merits and demerits.

(iii) Comment on statement "Hydrogen is a alternative future source of energy".

GLA University, Mathura

II-Mid Term Examination, 2011-12

Course: - B.Tech. II-Year, III-Sem.

Subject: - Data Structure And Algorithm **Univ. Roll No:-**

Time:-90 Minutes **Total Marks:-40**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Two** from Group B and **Any Two** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
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GROUP/SECTION-A(1x16)

Select the correct answer:-

(i) A binary tree whose every node has either zero or two children is called

a. Complete binary tree b. Binary search tree c.Extended binary tree d. None of above

(ii) The depth of a complete binary tree is given by

a. $D_n = n \log_2 n$ b. $D_n = n \log_2 n+1$ c. $D_n = \log_2 n$ d. $D_n = \log_2 n+1$

(iii) In a binary tree, certain null entries are replaced by special pointers which point to nodes higher in the tree for efficiency. These special pointers are called

a. Leaf b. branch c. path d. thread

(iv) The in order traversal of tree will yield a sorted listing of elements of tree in

a. Binary trees b. Binary search trees c. Heaps d. None of above

(v) If every node u in G is adjacent to every other node v in G, A graph is said to be

a. isolated b. complete c. finite d. strongly connected

Fill in the blanks:-

(vi) When inorder traversing a tree resulted E A C K F H D B G; the preorder traversal .....

(vii) In a heap, element with the greatest key is always in the _____ node.

(viii) A(n) _____ graph is a graph in which each vertex has a connection to every other vertex.

(ix) Infix notation of the prefix expression (-*/A**BCDE) is .....

(x) A node with no children is called a .....

(xi) Nodes that have degree.....is called terminal nodes

True And False:-

(xii) When a node is inserted in a binary tree it must be inserted as a leaf node.

(xiii) A Binary tree cannot have odd number of nodes.

(xiv) A tree can be traversed in non-recursive manner.

(xv) Sum of outdegree and indegree of a node is called its total degree

(xvi) A graph in which all the nodes are of equal degree is called complete graph

GROUP/SECTION-B(6X2) 6x2

(i) a. Suppose the following sequences list the node of a binary tree T in preorder and inorder respectively :

Preorder: G,B,Q,A,C,K,F,P,D,E,R,H

Inorder: Q,B,K,C,F,A,G,P,E,D,H,R

b. Explain the concept of threaded binary tree.

(ii) a. Draw the maxheap for the following list of elements

L=< 20,10,5,4,25,70,60,40 >

b. Write algorithm for inserting n element in a heap.

(iii) a. Create a Binary Search tree for the following list of elements:

L=< 5,4,10,20,13,15,40,30>

b. Write an algorithm for inserting an element in BST.

GROUP/SECTION-C(6X2)

(i) For the following graph find minimum spanning tree using prim's algorithm (Fig 1)

(ii) Write algorithm for deleting an element from Binary search Tree.

(iii) Given the characters S(a,b,c,d,e,f) with the following probability

P=<29,25,20,12,05,09>

Build a Huffman tree and also find Huffman code

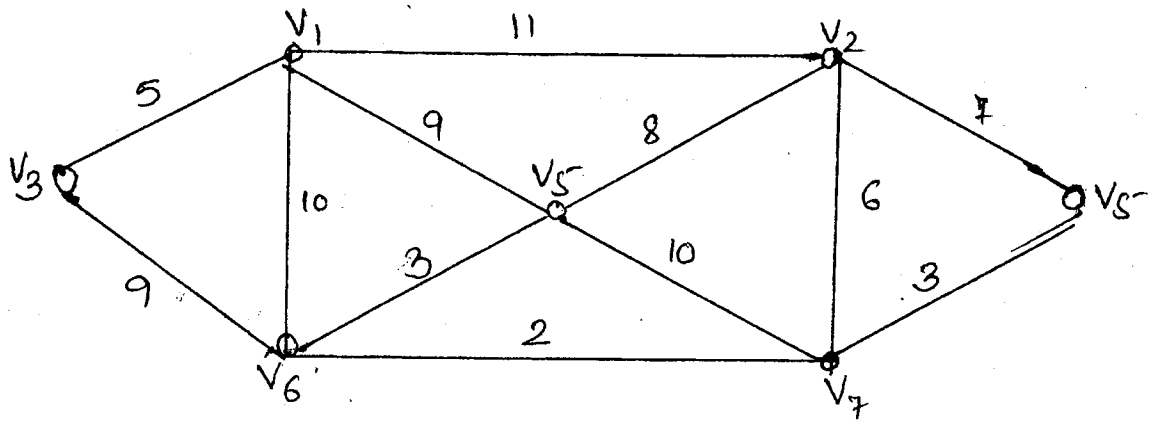


Fig 1.

GLA University, Mathura

II-Mid Term Examination, 2011-12

Course: - B.Tech (CS) II-Year, III-Sem.

Subject: -Computer Organization **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-40**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Two** from Group B and **Any Two** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

GROUP/SECTION-A

I Fill in the blanks **(16)**

a) A micro operation is an elementary operation performed on the information stored in one or more .....
b) Register that holds an address for the memory unit is usually called a .....
c) A bus system can be constructed with three state gates instead of .....
d) .....micro operations perform bit manipulation operation on non numeric data stored in registers.
e) The digital circuit that forms the arithmetic sum of two bits and the previous carry is called a .....
f) The circular shift also known as a ..... operation.
g) The control variable at any given time can be represented by a string of 1's and 0's called a .....
h) The ..... specifies one or more micro operation for a system.
i) The next address generator determines the address sequence that is read from control memory called a .....
j) The control memory is assumed to be a .....within which all control information is permanently stored.
k) When the Control signals are generated by hardware using conventional logic design techniques, the control unit is said to be .....
l) The control function that specifies a micro operation is a .....variable.
m) An ..... Is a group of bits that instruct the computer to perform a specific operation.
n) A..... is a set of instruction that specify the operation.
o) The process of saving the data and instructions permanently .....
p) The ALU and CU of a computer system are jointly known as.....

GROUP/SECTION-B

II Attempt any two **(12)**

Q1 Describe various type of micro operation. Show the hardware that implements the following statements. Include the logic gates for the control function and a block diagram for the binary counter with a count enable input.

$xyT_0 + T_1 + y'T_2 : AR \leftarrow AR+1$

Q2 Explain block diagram of Control unit and its components.

Q3 What is the difference between a microprocessor and a micro program? Is it possible to design a microprocessor without a micro program? Are all Micro programmed Computers also a Microprocessors.

GROUP/SECTION-C

III Attempt any two **(12)**

Q1 The control memory has 4096 words of 24 bit each.

a) How many bits are there in the control address register?
b) How many bits are there in each of the four inputs shown going into the multiplexers.
c) What are the number of inputs in each Multiplexer and how many multiplexers are needed.

Q2 Explain the difference between hardwired control and micro programmed control. Is it possible to have a hardwired control associated with a control memory.

Q3 Starting from a initial value of R=11011101, determine the sequence of binary values in R after a logical shift left, followed by a circular shift left, followed by a logical shift right and a circular shift left.

GLA University, Mathura

II-Mid Term Examination, 2011-12

Course: - B.Tech **Year:II** **Sem:III**

Subject: Object Oriented Programming

Uni. Roll No:-

Time:-90 Minutes **Total Marks:-40**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Two** from Group B and **Any Two** from Group C.Each Question of Group A carry 1 Marks
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

GROUP/SECTION-A

Objective Type Questions:

1. Static member function can be called using
   (a) Class name (b) Object name (c)same as Normal function (d) None of these
2. The return type of contructor is
   (a)void (b) int (c)char (d) Does not have return type
3. The process by which objects of one class acquire the properties of objects of another class is known as
   (a)Inheritence (b) Data Abstraction (c)Polymorphism (d)Data Hiding
4. Which symbol is used for preceding the destructor
   (a) ! (b) ~ (c) $ (d)*
5. The statement A obj=A(1,2) is
   (a)Explicit call of constructor (b)Implicit call of Constructor (c)Both (d)None
6. The keyword new is used for
   (a) creating new object (b) creating new class (c) allocating memory (d) None
7. The order of execution of constructors in the statement class A:public B , public C{};
   (a) A(),B(),C() (b) B(),C(),A() (c) C(),B(),A() (d) A(),C(),B()
8. Virtual function comes under
   (a)Compile Time Polymorphism (b)Run Time Polymorphism (c)Both (d) None
9. The default value of static data member is
   (a)1 (b)-1 (c)Garbage (d) 0
10. class A:B{}; In this declaration ,class B is inherited
    (a) publicly (b) privately (c)protectedly (d) None of these

Fill in the blanks:-

11. The keyword private is a _____ specifier.
12. Constructors should be declared in _____ part of class.
13. The _____ Constructor does not contain any parameter.

True or False:-

14. A friend function can be declared only in public part of the class.
15. The parameters of constructors can accept the reference to its own class .
16. Scope Resolution operator i.e. :: can be overloaded.

GROUP/SECTION-B

Q1. Write a short note on:-

(a) Constructor (3 Marks)

(b) Destructor (3 Marks)

Q2. (a) What is friend Function? Discuss it. (3 Marks)

(b) Write a C++ program which makes the use of friend function (3 Marks)

Q3. (a) When friend function is compulsory in case of operator overloading. (3 Marks)

(b) When the definition of default Constructor is compulsory (3 Marks)

GROUP/SECTION-C

Q1. Explain Inheritance and it's types. Give one example of Multiple Inheritance. (6 Marks)

Q2. Write a C++ program to add two complex numbers by overloading + operator .Here + is binary operator. (6 Marks)

Q3. Explain Virtual Function with the help of example. (6 Marks)